



Effects of perinatal stress on the anxiety-related behavior of the adolescence mouse

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Abstract

We evaluated the effects of sound noise or forced swim stress applied to pregnant mice or to neonatal mice on the anxiety-related behavior using the elevated plus-maze test performed during the age of 5 weeks. The forced swim stress applied at the late gestation period, days 10–18 of pregnancy, caused a significant reduction of the body weight gain of the dams. However, the anxiety-related behavior of the male and female offspring were not affected by the antenatal stress treatment. When the forced swim stress was applied to the neonatal mice during the late lactation period, 14–18 days after birth, the male mice showed an elevated level of anxiolytic behavior accompanying the reduction of the emotion-related motor activity. The anxiety-related behavior of the female mice was not affected by the stress treatment. Furthermore, we applied the sound noise or forced swim stress to the neonatal mice immediately after the weaning, 21–25 days after birth. The stress applied after the weaning period had no effect on the anxiety-related behavior. These results suggested that the stress applied during the lactation period, but not that during the antenatal period, nor after the weaning period, might have gender-dependently reduced the anxiety level of the male mouse. It was shown that the effects of perinatal stress on the anxiety-related behavior of the adolescent mouse varied according to the period of application and gender. The hypothesis that gender-dependent abnormalities in neurodevelopment might be caused by the excess stress applied to the breast-fed infant is of importance in elucidating the relationship between the psychoneurotic disorder in childhood and the environment stress of the breast-fed infant.

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It is generally accepted that antenatal and/or early postnatal stress adversely affects human development, increasing the susceptibility to diseases later in life (Barker, 1998, 2002), as well as altering behavioral and cognitive development (Ellis and Cole-Harding, 2001; O'Connor et al., 2002; Charmandari et al., 2003). These effects have been linked to persistent alterations in the hormonal and metabolic responses to stress associated with changes in the regulation of the hypothalamic–pituitary–adrenal (HPA) axis (Charmandari et al., 2003; Matthews, 2002). The mechanisms underlying these lasting developmental responses are not fully understood.

We have been studying the relationships between the application of perinatal stress and functional brain development from the perspective that the sexual differentiation of the brain is critical for its effect (Becu-Villalobos et al., 1997). We already demonstrated that the antenatal noise stimulation to

rats, combined with the forced swim stress, did not affect growth during the postnatal period, but adversely affected their emotional state and learning capacity (Nishio et al., 2001). The latter effect of the stress treatment was shown to be sensitive to anxiolytics, and was observed only in male offspring, but not in female. Maternal stress also caused an increase in the total number of errors by male offspring, but not female, in the water-maze test (Nishio et al., 2001). This is the first report that the gestational stress might affect the development of offspring in a gender selective manner. The ethological and neurochemical mechanisms for the stress effects have not been elucidated in detail. Ward (1972) hypothesized that the exposure of pregnant rats to environmental stressors modified the normal process of sexual behavior differentiation in male fetuses by decreasing the functional testosterone and elevating androstenedione levels during perinatal development.

There is a rich collection of behavioral pharmacological literature on rodent anxiety-like behavior that has been readily adapted to the study of the mice. Among them, the elevated plus-maze is now commonplace in neuroscience laboratories

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around the world. The elevated plus-maze is an animal model where the behavioral repertoire of rodents is used to detect effects on anxiety, and it has proved to be bidirectionally sensitive to manipulations of anxiety (Espejo, 1997; Pellow and File, 1986). Drugs with anxiolytic action in humans, such as diazepam, produce decreases in the avoidance of aversive/threatening areas in the mice (Cole and Rodgers, 1995; Tokumo et al., 2006). TFMPP, an anxiogenic drug with 5-HT_{2C} receptor agonist activity, has the opposite effect (Setem et al., 1999; Tokumo et al., 2006). Thus the elevated plus-maze provides a tool for the analysis of the ethological effect of the stress applied to the mouse on the anxiety-related behavior.

In the present study, we applied perinatal stresses to mice and examined their effects of the anxiety-related behavior of the adolescent mice. The effects of the sound noise or forced swim stress applied to pregnant mice or to neonatal mice on the anxiety-related behavior of adolescent mice in the elevated plus-maze test were evaluated.

1. Experimental procedures

1.1. Animals

Pregnant ddY mice were obtained from Japan SLC (Shizuoka Lab, Japan). The experimental animals received food (CE-2, Clea Japan, Inc.) and tap water ad libitum, and were housed under standard laboratory conditions (room temperature 23 ± 1 °C, humidity $55 \pm 5\%$, lights on 07:00–19:00). The Guidelines of Ethical Standards for Animal Experiments of Fukuyama University were followed.

1.2. Stress treatment

1.2.1. Antenatal stress treatment on the pregnant mice

Antenatal stress was applied to the dams during the late stage of pregnancy. On day 10 of pregnancy, the pregnant mice were divided into three groups, namely pregnant mice [dam-N], [dam-SO] and [dam-SW], and they were manipulated as described below. Five mice were used in each group. The [dam-SO] group was exposed to 90 dB noise every minute for 15 min once a day from day 10 to 18 of pregnancy. The neonatal male and female mice from the [dam-SO] were labeled [pre-SO-m] and [pre-SO-f], respectively. The [dam-SW] group was forced to swim for 15 min once a day from day 10 to 18 of pregnancy. The neonatal male and female mice from the [dam-SW] group were named [pre-SW-m] and [pre-SW-f], respectively. No stress manipulation was applied to the pregnant mice [dam-N] group. The neonatal male and female mice from [dam-N] were named [pre-N-m] and [pre-N-f], respectively. All neonatal mice were reared by each dam under standard laboratory conditions.

1.2.2. Postnatal stress treatment during the lactation period

Postnatal stress was applied to the neonatal mice during the late lactation period. At 14 days after birth, the neonatal male and female mice were divided into three groups, namely the neonatal mice [dl-N-m] or [dl-N-f], [dl-SO-m] or [dl-SO-f], and [dl-SW-m] or [dl-SW-f], and they were manipulated as described below. The neonatal male mice [dl-SO-m] and female mice [dl-SO-f] were exposed to the 90 dB noise every minute for 15 min once a day from 14 to 18 days after birth. The neonatal male mice [dl-SW-m] and female mice [dl-SW-f] were forced to swim for 15 min once a day from 14 to 18 days after birth. No stress manipulation was applied to the neonatal male mice [dl-N-m] and female mice [dl-N-f].

1.2.3. Postnatal stress treatment after the weaning period

Postnatal stress was applied to the neonatal mice after the weaning period. The neonatal mice were weaned at 21 days after birth. They were divided into

three groups, namely neonatal mice [aw-N-m] or [aw-N-f], [aw-SO-m] or [aw-SO-f], and [aw-SW-m] or [aw-SW-f], and they were manipulated as described below. The neonatal male mice [aw-SO-m] and female mice [aw-SO-f] were exposed to the sound noise of 90 dB every minute for 15 min once a day from 21 to 25 days after birth. The neonatal male mice [aw-SW-m] and female mice [aw-SW-f] were forced to swim for 15 min once a day from 21 to 25 days after birth. No stress manipulation was applied to the neonatal male mice [aw-N-m] and female mice [aw-N-f].

1.3. Elevated plus-maze test

The anxiety-related behaviors of the mice were studied using an elevated plus-maze test as described previously (Tokumo et al., 2006). Briefly, the mouse was brought individually to the testing room 1 h before the test was started. The elevated plus-maze apparatus was made of wood-block and consisted of four arms (39.5 cm long $\times$ 10 cm wide) fixed to a central platform (10 cm $\times$ 10 cm): two had 12 cm high walls (closed-arms) and two had no border in place of the walls (open-arms). The maze was elevated to a height of 40 cm. For testing, the mouse was placed individually in the center of the maze, its head facing an open-arm (the same for all mice). An entry in a given arm was counted when the mouse had completely entered the arm. The primary measures were the number of entries and the time spent in each arm. These measures enabled the computation of the number of entries to the open- and closed-arms, and the time spent on the open- and on closed-arms, respectively. To balance for potential locomotion biases, the number of entries and time on the open-arms was expressed as a percentage of the total number of entries to all arms, or of the time spent in all arms, respectively ($(\text{open}/[\text{close} + \text{open}]) \times 100$). The absolute number of closed-arm entries was measured to dissociate the effect of the treatment on anxiety and general motor activity. The elevated plus-maze testing lasted for 5 min and was videotaped for analysis. For the evaluation of anxiety-related animal behavior, the percentage of time spent on open/total arms and the percentage of open/total arm entries were measured as conventional anxiety indices.

1.4. Statistical analysis

For the analysis of behavioral data, repeated measure one-way ANOVAs and Student's *t*-tests for independent samples were used. The *p*-values given are two-tailed, and represent measures of effect.

2. Results

2.1. Antenatal stress

2.1.1. Effects of the stress applied to pregnant mice on their body weight gain

Fig. 1 illustrates the cumulative body weight gain of the dams, [dam-N], [dam-SW] and [dam-SW]. Significant inhibitory effects of the stress treatment on the body weight gain of the dams were noted. The body weight gain of [dam-SO] showed a tendency to decrease, and that of [dam-SW] was significantly less than that of [dam-N]. The forced swim stress applied in the present study was strong enough to cause changes in one of the physical parameters of the pregnant mice.

2.1.2. Effects of antenatal stress on the anxiety-related behavior in the elevated plus-maze test

The dams that were manipulated with the noise or forced swim stress gave birth normally, as did the non-stressed dams. All neonatal mice from each group of dams grew normally to all appearances, and they were weaned on postnatal day 21. The body weights of the male or female mice 5 weeks after birth were as follows: “male” [pre-N-m] 28.9 ± 3.7 g ($n = 12$),

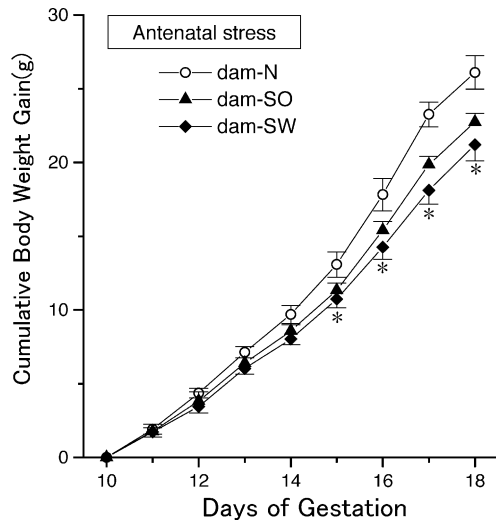


Fig. 1. Effects of the sound noise or forced swim stress on the cumulative body weight gain of the dams. The sound noise stress [dam-SO] or the forced swim stress [dam-SW] was applied from day 10 to 18 of gestation, as described in the text. The group [dam-N] was manipulated without the stress applications. Data are from five animals in each group. Significance: * $P < 0.05$ (vs. [dam-N]).

[pre-SO-m] 31.6 ± 1.3 g ($n = 12$), [pre-SW-m] 29.2 ± 1.6 g ($n = 12$), “female” [pre-N-f] 25.9 ± 3.0 g ($n = 12$), [pre-SO-f] 25.3 ± 1.9 g ($n = 12$), [pre-SW-f] 25.5 ± 2.1 g ($n = 12$). Among the groups, significant differences were not observed in the body weights of the male and female neonatal mice. At 5 weeks after birth, the mouse was subjected to the elevated plus-maze test to evaluate the anxiety-related behavior. The female mice [pre-SO-f] had a tendency to spend an increased percentage of time in the open-arms, an anxiolytic index of the animals, but the changes were not statistically significantly different from those observed in [pre-N-f] (Fig. 2a). Further-

more, the absolute number of entries to the closed-arm, an index of the locomotor activity in the maze, showed a tendency to decrease in the stressed female mice, [pre-SO-f] and [pre-SW-f], but the change was not significantly different from that observed in [pre-N-f] (Fig. 2c). Thus, the stress applied during the late gestation period had no significant effect on the anxiety-related behavior and the locomotor activity of the offspring.

2.2. Postnatal stress treatment during the lactation period

2.2.1. Effects of the stress applied to the neonatal mouse during the lactation period on their body weight gain

The sound noise or forced swim stress was applied to the neonatal mice in the late lactation period, from 14 to 18 days after birth. Fig. 3 illustrates the effects of stress treatment on the cumulative body weight gain during the stress application. The body weight gain of the female mice [dl-SO-f] was slightly reduced compared to that of [dl-N-f] (Fig. 3). The body weight gain of the male mouse was not affected by the stress manipulations (Fig. 3).

2.2.2. Effects of postnatal stress during the lactation period on the anxiety-related behavior in the elevated plus-maze test

The anxiety-related behaviors of the neonatal mice that were manipulated with stress during the late lactation period were evaluated. After the sound noise or forced swim stress application, the mice were weaned on postnatal day 21 under normal conditions. The body weights of the male or female mice 5 weeks after birth were as follows: “male”, [dl-N-m] 29.1 ± 1.7 g ($n = 20$), [dl-SO-m] 28.9 ± 2.8 g ($n = 17$), [dl-SW-m] 28.9 ± 1.7 g ($n = 17$), “female”; [dl-N-f] 26.0 ± 1.1 g ($n = 22$), [dl-SO-f] 23.9 ± 1.7 g ($n = 17$), [dl-SW-f] 25.2 ± 1.7 g ($n = 16$). The body weights of a group of female mice

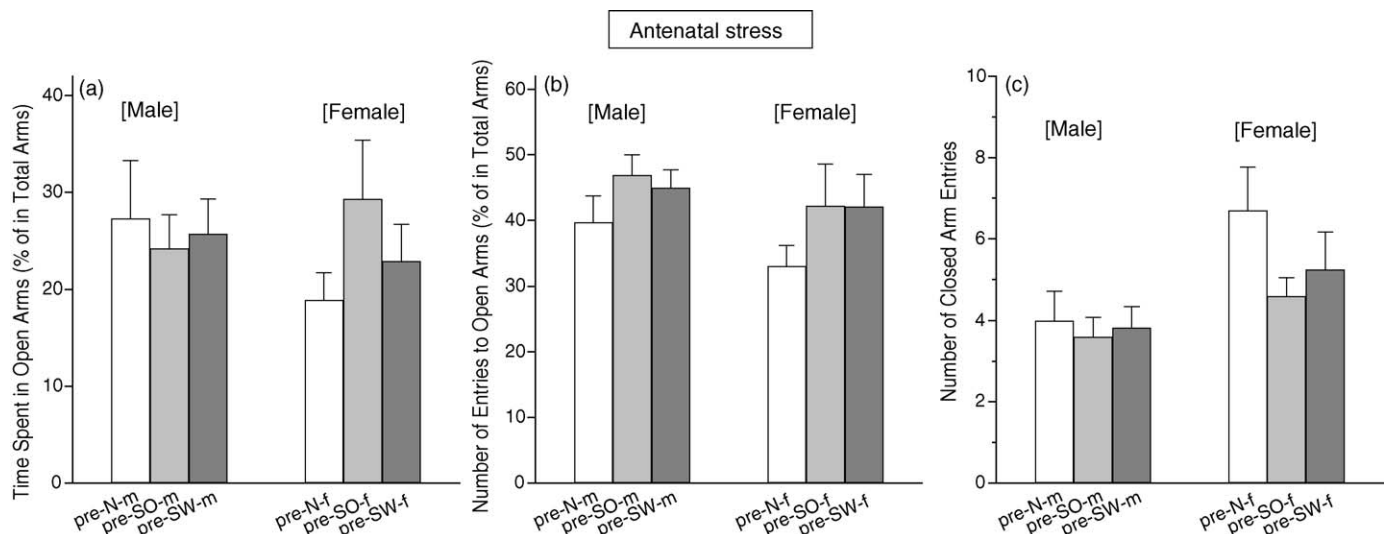


Fig. 2. The effects of the sound noise or forced swim applied during the antenatal period on the behavioral parameters of offspring in the elevated plus-maze: (a) time spent in open-arms (% of in total arms), (b) number of entries to open-arms (% of in total arms) and (c) number of closed-arm entries. The male neonatal mice, [pre-N-m], [pre-SO-m] and [pre-SW-m], were delivered from the dam, [dam-N], [dam-SO] and [dam-SW], respectively. The female neonatal mice, [pre-N-f], [pre-SO-f] and [pre-SW-f], were delivered from the dam, [dam-N], [dam-SO] and [dam-SW], respectively. The mice were subjected to the elevated plus-maze test at an age of 5 weeks after birth. Data are expressed as the mean value $\pm$ S.E.M. Data are from 12 animals in each group.

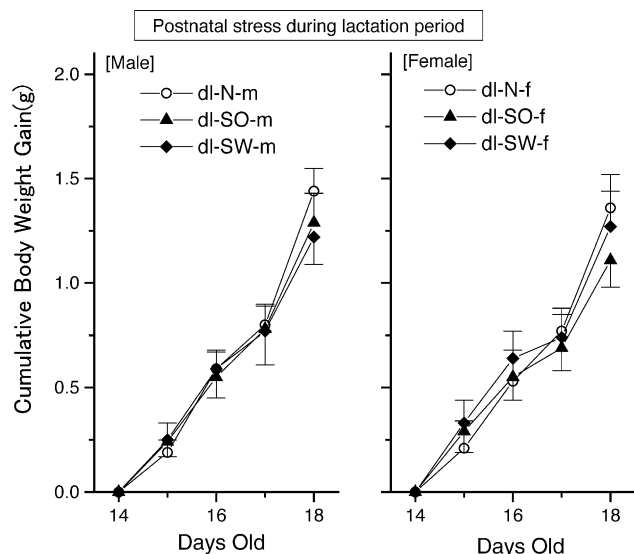


Fig. 3. Effects of the sound noise or forced swim stress applied during the lactation period on the cumulative body weight gain of the neonatal mice. The sound noise stress [dl-SO-m], [dl-SO-f] or the forced swimming stress [dl-SW-m], [dl-SW-f] was applied to male and female mice from day 14 to 18 after birth, as described in the text. Groups [dl-N-m] and [dl-N-f] were manipulated without the stress applications. Data are from 16 to 20 animals in each group.

those received the sound noise stress, [dl-SO-f], were significantly reduced compared to that of [dl-N-f]. No changes were observed in the body weights of the other stressed groups of male or female neonatal mice. In the elevated plus-maze test performed 5 weeks after birth, the male mice, [dl-SW-m], had a tendency to increase the percentage of time spent in the open-arms (Fig. 3a). A significant increase in the percentage of entries to open-arms was observed in the male mice that were manipulated with the forced swim stress [dl-SW-m] (Fig. 3b). Furthermore, the absolute number of entries to the closed-arm

significantly decreased in the forced swim stressed male mice [dl-SW-m] (Fig. 3c). Thus, it was indicated that the forced swim stress applied to male mice during the lactation period caused the reduction of the anxiety level accompanying the reduction of locomotor activity in the maze. No such stress effects were observed in the female mice (Fig. 4a–c).

2.3. Postnatal stress treatment after weaning period

2.3.1. Effects of the stress applied to the neonatal mouse after the weaning period on their body weight gain

The sound noise or forced swim stress was applied to the mice immediately after the weaning period, from 21 to 25 days after birth. Fig. 5 illustrates the effects of the stress treatment on the cumulative body weight gain during the stress application. The body weight gain of the neonatal mice did not change among the groups.

2.3.2. Effects of postnatal stress after the weaning period on the anxiety-related behavior in the elevated plus-maze test

The anxiety-related behavior of the neonatal mice that were manipulated with the sound noise or forced swim stress immediately after the weaning period was evaluated. After the stress application, the mice were raised under normal conditions. The body weights of the male or female mice 5 weeks after birth were as follows: “male”, [aw-N-m] 32.1 ± 0.4 g ($n = 16$), [aw-SO-m] 31.4 ± 0.5 g ($n = 16$), [aw-SW-m] 32.0 ± 0.7 g ($n = 16$), “female” [aw-N-f] 27.1 ± 0.3 g ($n = 16$), [aw-SO-f] 26.5 ± 0.3 g ($n = 16$), [aw-SW-f] 27.6 ± 0.4 g ($n = 16$). Among the groups, no significant differences in the body weights of the male or female neonatal mice among the groups was observed. The stress manipulation applied to the male and female mice immediately after weaning did not affect the percentage of time spent in the open-arms, or

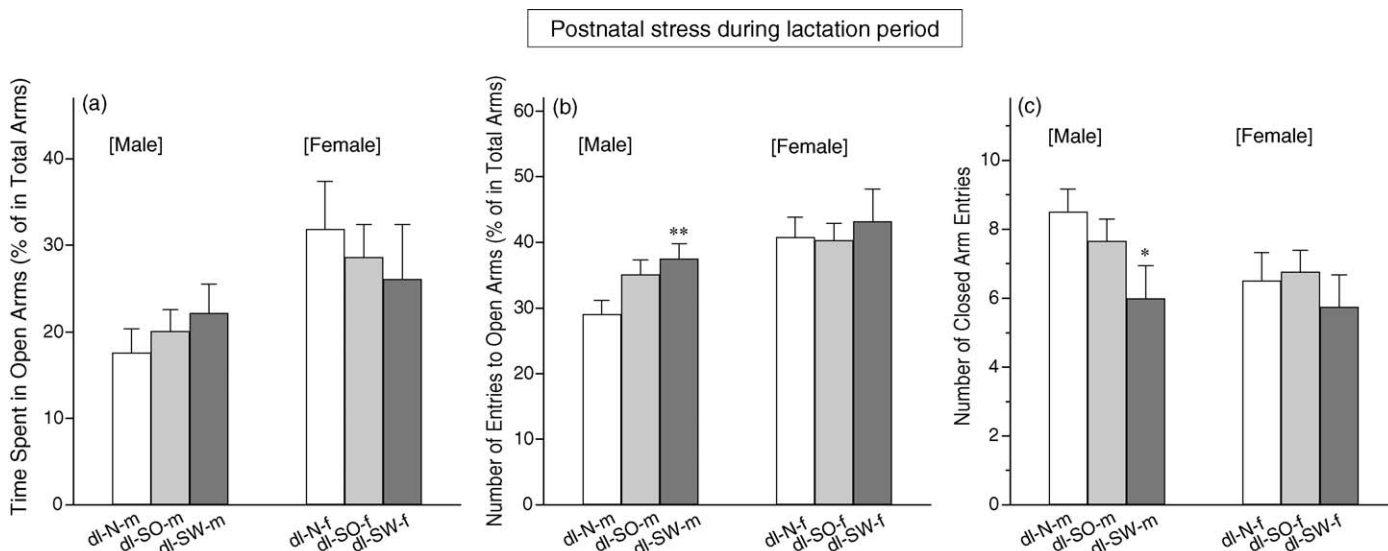


Fig. 4. The effects of the sound noise or forced swim stress applied during the lactation period on the behavioral parameters in the elevated plus-maze: (a) time spent in open-arms (% of in total arms), (b) number of entries to open-arms (% of in total arms) and (c) number of closed-arm entries. The sound noise stress [dl-SO-m], [dl-SO-f] or the forced swimming stress [dl-SW-m], [dl-SW-f] was applied to the male and female mice from day 14 to 18 after birth, as described in the text. Groups [dl-N-m] and [dl-N-f] were manipulated without the stress applications. The mice were subjected to the elevated plus-maze test at the age of 5 weeks after birth. Data are expressed as the mean value $\pm$ S.E.M. Data are from 16 to 20 animals in each group. Significance: * $P < 0.05$ (vs. [dl-N-m]), ** $P < 0.01$ (vs. [dl-N-m]).

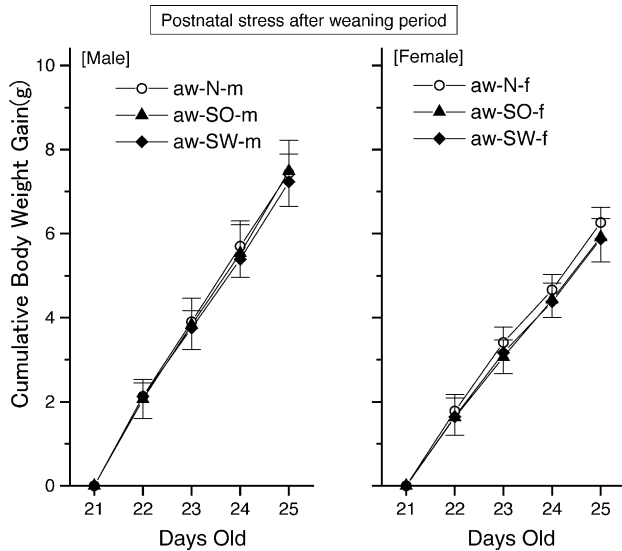


Fig. 5. Effects of the sound noise or forced swim stress applied immediately after the weaning on the cumulative body weight gain of the neonatal mice. The sound noise stress [aw-SO-m], [aw-SO-f] or the forced swimming stress [aw-SW-m], [aw-SW-f] was applied to male and female mice from day 21 to 25 after birth, as described in the text. The group [aw-N-m] and [aw-N-f] were manipulated without the stress applications. Data are from 16 animals in each group.

the percentage of entries to the open-arms at 5 weeks of age (Fig. 6a and b). The locomotor activity of the groups of stressed female mice, [aw-SO-f] and [aw-SW-f], in the maze showed a tendency to be increased by the manipulation, but the changes were not significant from that of [aw-N-f] (Fig. 6c).

3. Discussion

In the present study, we examined the effect of perinatal stress on the anxiety-related behavior, one of the parameters of

the brain functional development, using the mouse as the experimental animal. Thus, we evaluated the effect of the sound noise or forced swim stress applied to the antenatal mouse or to the neonatal mouse. The anxiety-related behavior of the mice was evaluated in the elevated plus-maze test at an age of 5 weeks. In the first experiment, the effect of stress applied during the antenatal period was analyzed. The forced swim stress applied during the late period of gestation caused a significant reduction of the body weight gain of the dams. The anxiety-related behavior of the male and female offspring at an age of 5 weeks, however, was not affected by the stress treatment. Thus, it was found that the intrauterine stress had no effect on the emotional behavior of the male and female mice in their growth period after birth. These results did not concur with those observed in our previous report obtained using rats as an experimental animal (Nishio et al., 2001). We reported that the sound noise and forced swimming stress, applied to pregnant rats, induced gender-dependent effects on the emotional behavior and learning ability of the neonatal rats. Thus, antenatal stress potentiated the sound-induced loss of locomotor activity, referred to as emotional behavior, in the male offspring, but not in the female offspring. These disparities between the mouse and rat might be explained by the suggestion that the critical period of sexual differentiation in the brain development is dissimilar between these animals (Arai, 1990). It was reported that the critical period of sexual differentiation in the mouse brain development appeared in the postnatal period, later than in the rat brain development (Arai, 1990).

Conforming with these hypotheses, we evaluated the effects of postnatal stress to observe the possible gender-dependent influences on mouse brain development. We applied sound noise or forced swim stress to the neonatal mouse during the late lactation period, 14–18 days after birth. In the elevated

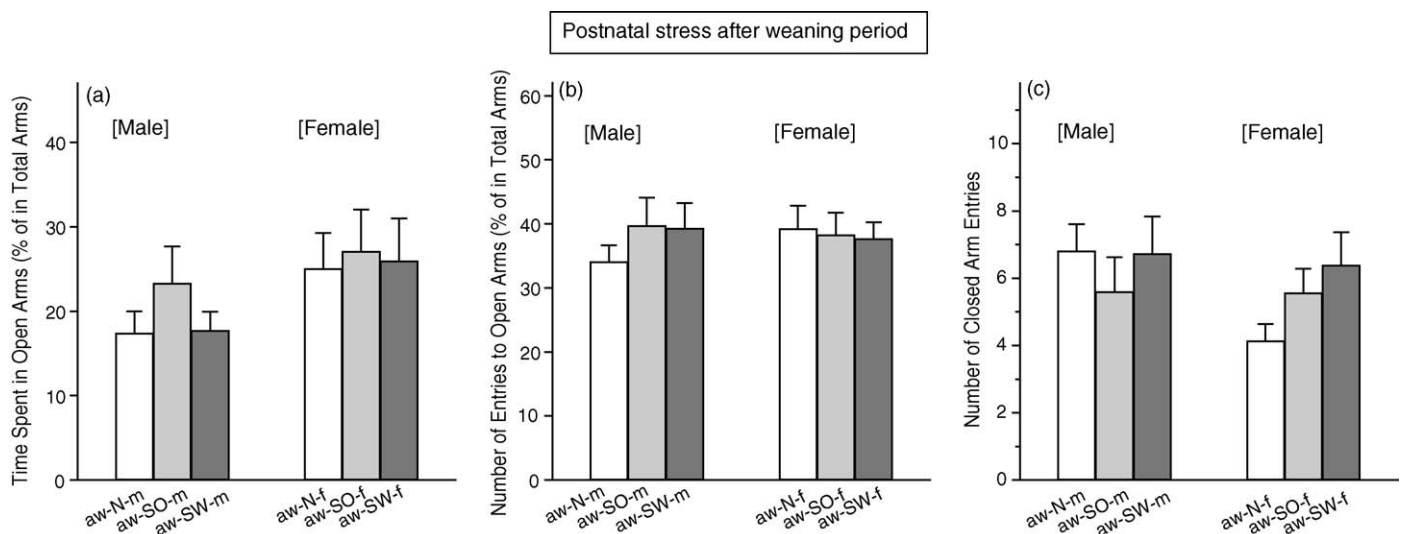


Fig. 6. The effects of the sound noise or forced swim stress immediately after the weaning on the behavioral parameters in the elevated plus-maze: (a) time spent in open-arms (% of in total arms), (b) number of entries to open-arms (% of in total arms) and (c) number of closed-arm entries. The sound noise stress [aw-SO-m], [aw-SO-f] or the forced swimming stress [aw-SW-m], [aw-SW-f] was applied to male and female mice from day 21 to 25 after birth, as described in the text. The group [aw-N-m] and [aw-N-f] were manipulated without the stress applications. The mice were subjected to the elevated plus-maze test at an age of 5 weeks after birth. Data are expressed as the mean value $\pm$ S.E.M. Data are from 16 animals in each group.

plus-maze test performed at 5 weeks after birth, the male mice with the forced swim stressed revealed changes in their anxiety indices. The stressed male mice, [dl-SW-m], showed a significantly elevated level of anxiolytic behavior; that is, the number of entries to the open-arms increased in this group, with a tendency of increase in the percentage of time spent in the open-arms. Furthermore, the absolute number of closed-arm entries, an index of the locomotor activity in the maze, also increased in the stressed male mice, [dl-SW-m]. These results imply that the stress applied during the lactation period of the neonatal mouse might have reduced their anxiety level accompanied by the reduction of the emotion-related motor activity. These effects of the stress were not observed when the female neonatal mice were tested. Thus, it was found that the stress applied during the lactation period, thought to be the critical period of mouse brain development, caused a gender-dependent influence in the subsequent growth processes. These results were comparable to those that we found regarding the effects of antenatal stress applied to the pregnant rats on the emotional behavior and learning ability of the offspring. We should investigate the effects of stress applied during the critical period of mouse brain development on the emotional and the learning behavior in more detail, as well as on the neurochemical parameters of the developing brain.

For the final experiment, we applied the sound noise or forced swim stress to the neonatal mouse immediately after weaning, 21–25 days after birth. The elevated plus-maze test of the neonatal mice was performed 5 weeks after birth. The results showed that the stress applied after the weaning period had no effect on the anxiety-related behavior of the neonatal mouse in early adolescence. Thus, it was revealed that the gender-dependent effects of perinatal stress on the anxiety-related behavior of the early adolescent mouse might depend on the period of stress application.

In conclusion, it was found that the postnatal stress during the lactation period exercised a gender-dependent effect on the anxiety-related behavior of the mice when observed 5 weeks after birth, which is comparable to early adolescence in human growth. It is an interesting fact that a minor difference in the age of stress application, 2 weeks or 3 weeks after birth, caused significant differences in the effects on the anxiety-related behavior of the male mice in early adolescence. The hypothesis that gender-dependent neurodevelopmental abnormalities of function might be caused by excess stress applied to the breast-fed infants is of significance in elucidating the relationship

between the psychoneurotic disorders in the childhood and the environment stress of the breast-fed infant. The ethological and neurochemical significance of the present results, that the anxiety level of the 5-week-old male mice was reduced by the stress applied during the late lactation period, should be examined in more detail.

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